

Cybersecurity Training Diary

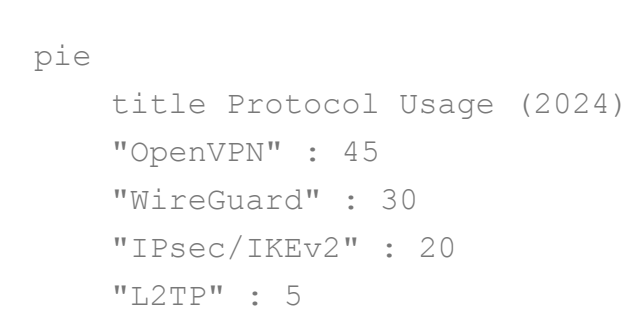
Day 9: VPNs & Secure Remote Access

1. Topics Covered

✓ VPN Fundamentals

Type	How It Works	Use Case	Security Risk
Site-to-Site	Connects entire networks	Branch office connectivity	Misconfigured tunneling
Remote Access	Individual user connection	Work-from-home	Credential theft
SSL/TLS	Browser-based (no client)	Contractors	Web app vulnerabilities
IPsec	Encrypts at OSI Layer 3	Enterprise networks	Complex configuration

✓ VPN Protocols Comparison



✓ Zero Trust Alternative

- **Beyond VPN:** "Never trust, always verify"
- **Components:**
 - Identity verification (MFA)
 - Device health checks
 - Micro-segmentation

2. Key Takeaways

WireGuard is faster than OpenVPN (modern cryptography)

Split tunneling improves speed but reduces security

Logging policies matter (ProtonVPN vs free VPNs)

3. Practical Exercises

Exercise 1: Set Up WireGuard VPN (Linux)

```
# Install
sudo apt install wireguard

# Generate keys
wg genkey | tee privatekey | wg pubkey > publickey

# Config template
echo "[Interface]
PrivateKey = <YOUR_PRIVATE_KEY>
Address = 10.8.0.1/24
ListenPort = 51820" > /etc/wireguard/wg0.conf
```

Exercise 2: Analyze VPN Traffic

1. Connect to any VPN
2. Run `tcpdump -i tun0`
3. Observe encrypted packets

4. Questions for Reflection

Why would a company choose IPsec over OpenVPN?

How could a hacker exploit a poorly configured VPN?

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